DATA ANALYSIS PROJECT (PANDAS)

March 24, 2026 (Piyush Patil)

OVERVIEW

1. Project background and description

With the increasing use of data in businesses, data analysis has become important for better decision-making. In this project, sales data is analyzed using Python and the Pandas library. The dataset is loaded, cleaned, and processed by performing operations like filtering, grouping, and summarizing. The aim is to extract meaningful insights and understand sales performance using simple data analysis techniques.

2. Project scope

This project can be extended to analyze larger and real-time datasets for better business insights. It can be used in various domains like retail, e-commerce, and sales management to track performance and trends. Advanced features like dashboards, predictive analysis, and automation can also be added in the future.

3. Objective

- To analyze sales data using Python Pandas, perform data cleaning, and generate useful insights.

4. Deliverables

- Data analysis code implemented using Python and Pandas
- Dataset used for analysis
- Visualizations representing insights
- Detailed project report with explanation and results

5. Dataset Used

The dataset used is a sample sales dataset containing product, category, sales, quantity, and city information.

6. Code

```
# Import Libraries
import pandas as pd
import matplotlib.pyplot as plt

# Load Dataset -- check the path of sales_data.csv file into your pc correctly and give path accordingly
df = pd.read_csv(r'C:\Users\Admin\OneDrive\Desktop\BCA\TY,BCA\InternSpark Internship\Task1-Data Analysis Project (Pandas)\sales_data.csv')

# Show Data
print("First 5 rows:")
print(df.head())

# Check Missing Values
print("\nMissing Values:")
print(df.isnull().sum())

# Data Cleaning
df = df.dropna()
df = df.drop_duplicates()

# Summary Statistics
print("\nSummary Statistics:")
print(df.describe())

# Grouping: Total Sales by Category
category_sales = df.groupby('Category')['Sales'].sum()
print("\nTotal Sales by Category:")
print(category_sales)

# Grouping: Total Sales by City
city_sales = df.groupby('City')['Sales'].sum()
print("\nTotal Sales by City:")
print(city_sales)

# Filtering: Sales greater than 500
high_sales = df[df['Sales'] > 500]
print("\nHigh Sales Data:")
print(high_sales)
```

```
# Plot Graph
category_sales.plot(kind='bar', title='Sales by Category')
plt.xlabel('Category')
plt.ylabel('Sales')
plt.show()
```

7. Steps Performed

- Loaded dataset using Pandas
- Checked and removed missing values
- Removed duplicate records
- Generated summary statistics
- Grouped data by category and city
- Filtered high sales records
- Visualized data using bar chart

8. Insights

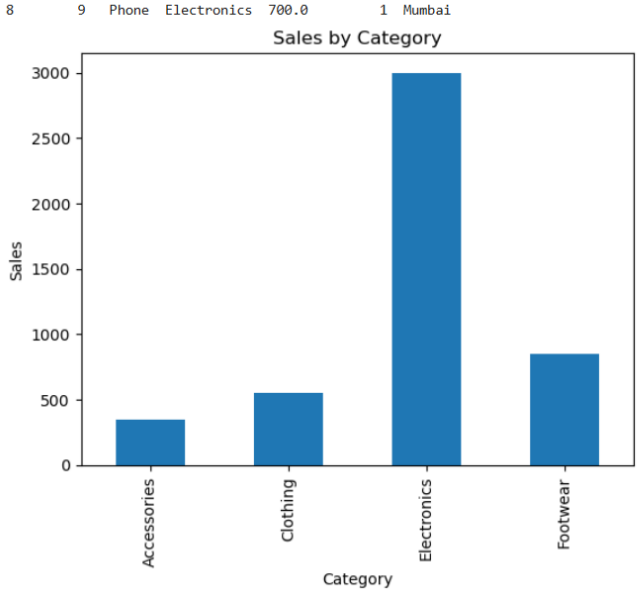
- *Electronics category has highest sales*
- *Mumbai shows strong sales performance*
- *High-value transactions are mostly from electronics products*
- *Data cleaning improves accuracy*

CODE, OUTPUT SCREENSHOTS :

```
•[9]: import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv(r'C:\Users\Admin\OneDrive\Desktop\BCA\TY,BCA\InternSpark Internship\Task1-Data Analysis Project (Pandas)\sales_data.csv')

print("First 5 rows:")
print(df.head())

print("\nMissing Values:")
print(df.isnull().sum())

df = df.dropna()
df = df.drop_duplicates()

print("\nSummary Statistics:")
print(df.describe())

category_sales = df.groupby('Category')['Sales'].sum()
print("\nTotal Sales by Category:")
print(category_sales)

city_sales = df.groupby('City')['Sales'].sum()
print("\nTotal Sales by City:")
print(city_sales)

high_sales = df[df['Sales'] > 500]
print("\nHigh Sales Data:")
print(high_sales)

category_sales.plot(kind='bar', title='Sales by Category')
plt.xlabel('Category')
plt.ylabel('Sales')
plt.show()
```

```
First 5 rows:
   OrderID Product     Category  Sales  Quantity    City
0        1  Laptop  Electronics  800.0         1  Mumbai
1        2   Shirt     Clothing  300.0         2    Pune
2        3   Phone  Electronics  600.0         1   Delhi
3        4   Shoes     Footwear  400.0         2  Mumbai
4        5   Watch  Accessories  200.0         1    Pune

Missing Values:
OrderID     0
Product     0
Category    0
Sales       1
Quantity    0
City        0
dtype: int64

Summary Statistics:
         OrderID      Sales   Quantity
count  10.00000   10.00000  10.000000
mean    5.50000  475.00000   1.500000
std     3.02765  262.73138   0.707107
min     1.00000  150.00000   1.000000
25%     3.25000  262.50000   1.000000
50%     5.50000  425.00000   1.000000
75%     7.75000  675.00000   2.000000
max    10.00000  900.00000   3.000000

Total Sales by Category:
Category
Accessories     350.0
Clothing        550.0
Electronics    3000.0
Footwear        850.0
Name: Sales, dtype: float64
```

```
Total Sales by City:
City
Delhi     1650.0
Mumbai    2150.0
Pune       950.0
Name: Sales, dtype: float64

High Sales Data:
   OrderID Product     Category  Sales  Quantity    City
0        1  Laptop  Electronics  800.0         1  Mumbai
2        3   Phone  Electronics  600.0         1   Delhi
5        6  Laptop  Electronics  900.0         1   Delhi
8        9   Phone  Electronics  700.0         1  Mumbai
```



9. Conclusion

In this project, sales data was successfully analyzed using Python and Pandas. Data cleaning, filtering, and grouping techniques were applied to extract useful insights. The results helped in understanding sales trends and performance across different categories and cities. This project demonstrates how data analysis can support better decision-making in real-world scenarios.